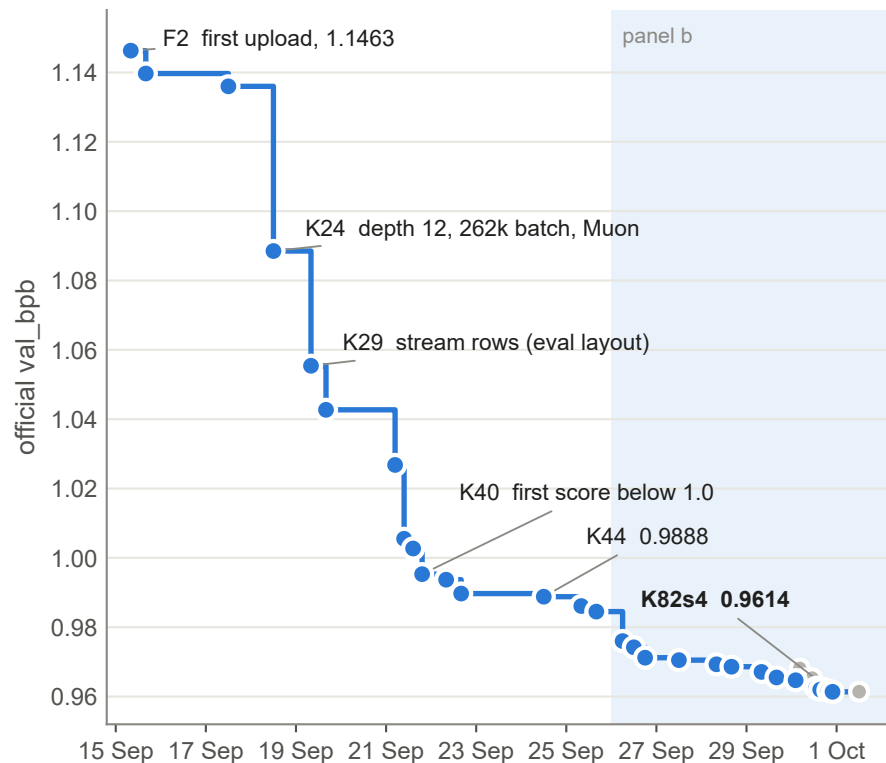
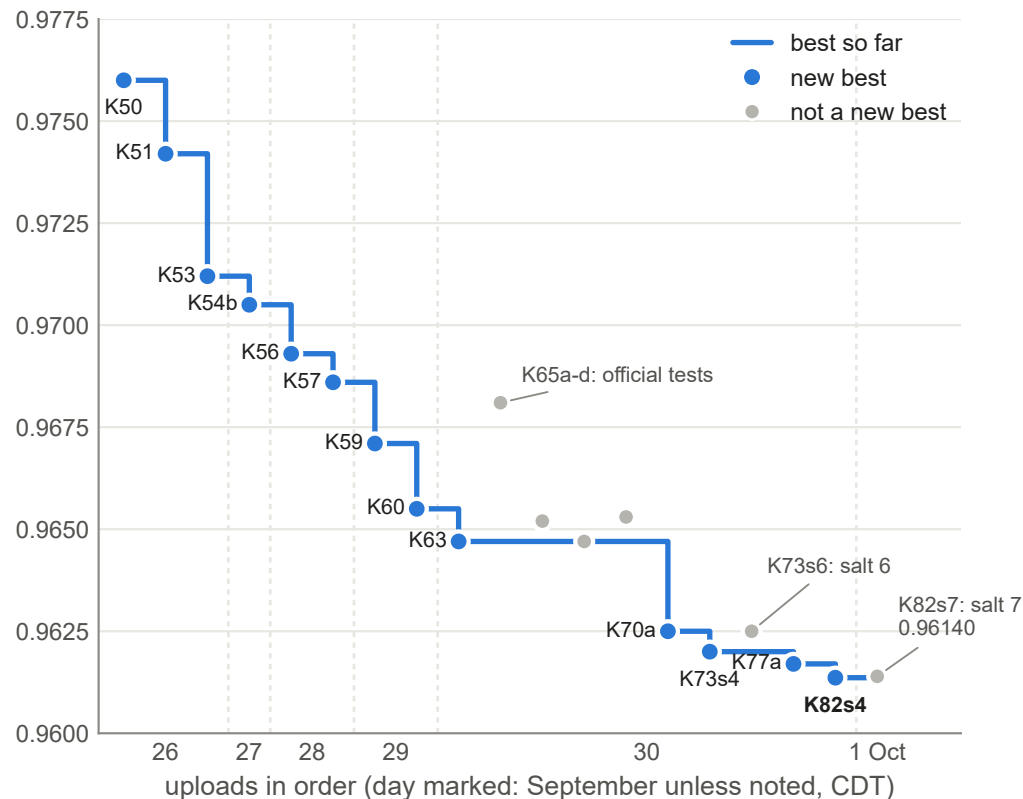


**a The whole campaign**



**b From K50, in upload order: each step is one change**



**What each new best in panel b changed**

**K50** batch warm-up, multi-token prediction, Muon schedule

**K51** key offset, accumulation schedule

**K53** ZeRO Muon, EMA prewarm, EMA every 32

**K54b** ZeRO-2 + AdamW ZeRO, 3-phase batch ramp

**K56** key-offset radius 0.3, accumulation in the graph

**K57** seed 73

**K59** LeakyReLU(0.35)<sup>2</sup> MLP

**K60** attention-source reuse

**K63** EMA off, cooldown 0.7, small-batch LR

**K70a** row pool 256 (−0.0022 official)

**K73s4** optimizer fusion, XU 48, c\_proj LR 0.8, salt 4

**K77a** EMA blend 0.6

**K82s4** EMA prewarm: best official, 0.96136